Hiring and Firing

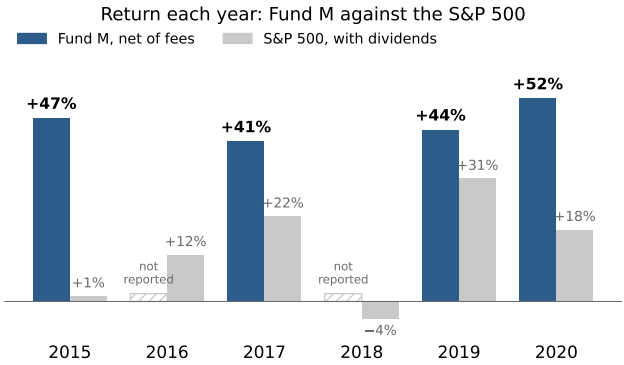
Strategies, managers, and one decision problem

Alan Moreira

NYU Stern — Data-Driven Investing

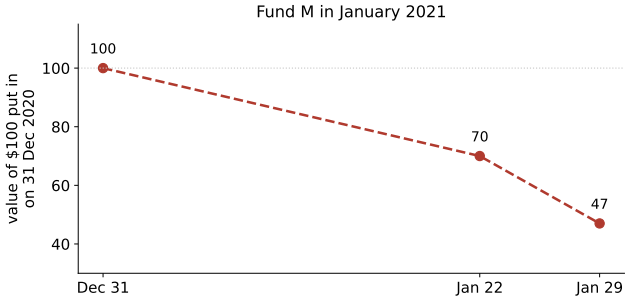
Fall 2026

Would you invest with this manager?



- About 30% a year, 2014–2020
- The S&P 500 over 2015–2020: about 13% a year
- +52% in 2020
- $12.5 billion to manage by January 2021
- Long-short US equity, run by a star stock-picker

January 2021: you are an investor. What do you do?



−53%
in one month

A Take your money out

B Stay

C Put more in

Two of the best-informed investors in the business chose C

WALL STREET JOURNAL Jan 24, 2021

A top hedge fund is down about 30% this year on losing short bets

REUTERS Jan 25, 2021

Citadel and Point72 put **$2.75 billion** into the fund

BENZINGA Jan 25, 2021

Citadel and partners: $2 billion. Steve Cohen's Point72: $750 million

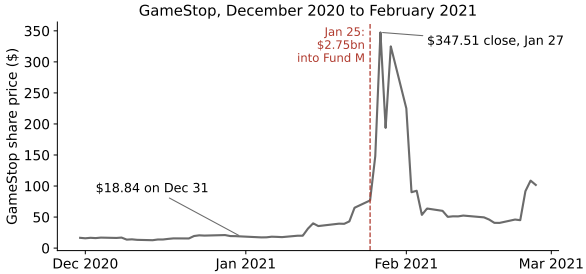
WALL STREET JOURNAL Jan 25, 2021

In return: money in the fund, plus a share of the firm's revenue

They put money in during the worst week. Why?

Fund M is Melvin Capital, and the trade was GameStop

- Founded in 2014 by Gabe Plotkin, one of Steve Cohen's top portfolio managers at SAC Capital
- Melvin was short GameStop
- Retail traders on Reddit's r/wallstreetbets bid the stock up 18-fold in four weeks
- Jan 28: Robinhood restricts buying of GameStop



REUTERS Feb 1, 2021

Melvin lost 53% in January. It ends the month with about $8 billion, new money included

What did Citadel and Point72 know that you did not?

- Cohen had watched Plotkin's process up close for years at SAC
- Point72 already had money in Melvin before January
- To them, the loss looked like one crowded short squeezed by retail buying, not a failed process
- They also bought a share of Melvin's revenue: a stake in the business, not only in the fund

If you trust the manager, January says little about his skill, and it may be a good time to get in

Same month, two readings

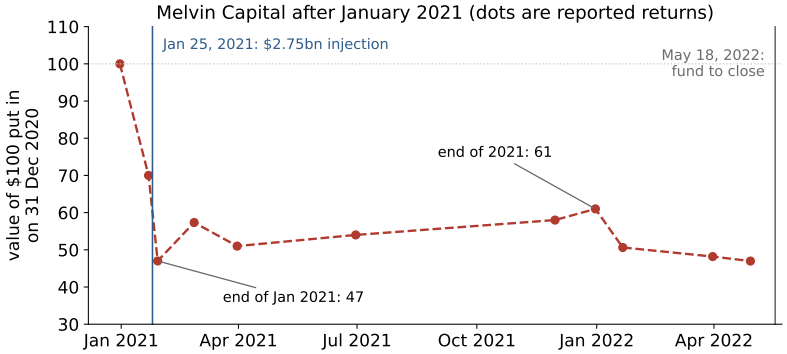
Suppose a month like January is **three times** likelier from an unskilled manager than from a skilled one.

	odds that he is skilled		
	before January	after January	what you do
Citadel, Point72	9 to 1 (90%)	3 to 1 (75%)	put money in
an investor with no inside view	1 to 1 (50%)	1 to 3 (25%)	take money out

- The same news divides both odds by three. Where you end up depends on where you started.

The formal version

What happened next



- Money put in at the end of January gained about 30% over the rest of 2021. Money already in lost 39% for the year.
- Citadel and Point72 took their money out in 2021–22. Melvin closed in May 2022. Sources

The question behind every hire and every fire

- Going in and getting out are statistical decisions
- How far bad news moves you depends on what you knew before it arrived
- One outcome does not grade the decision: Melvin closed, yet money that came in after the January crash gained about 30% over the rest of 2021

Same returns, different priors, opposite decisions

Going into a strategy is a statistical decision

You never see the true appraisal ratio. You see an estimate, from T years of data. Only unbiased data counts (Test–not EST/TUNE!)

$$\text{go in when} \qquad \widehat{AR}_T \;\geq\; a^* + \frac{z_p}{\sqrt{T}}$$

- a^*: your hurdle
- z_p: how sure you need to be (1.64 for 95%)
- $1/\sqrt{T}$: the standard error of the estimate
- This decision rule implies 95% of the strategies you got into will have appraisal higher than the hurdle

Walking away is the mirror image

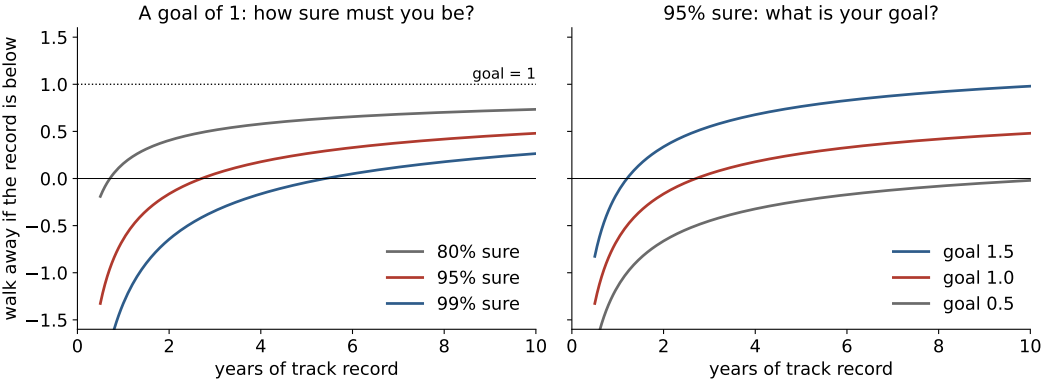
Give up once you are sure enough that the strategy does *not* meet your goal.

$$\text{walk away when} \qquad \widehat{AR}_T \;\leq\; a^* - \frac{z_p}{\sqrt{T}}$$

The same three ingredients: a **goal**, a **confidence**, and **time**

- In between the two lines, you do not know yet

The walk-away line



- More confidence means more patience. A higher goal lifts the whole line.

Three levers

Your goal is an appraisal ratio of 1. Walk away if the record is below:

years of record	80% sure	95% sure	99% sure
1	0.16	-0.64	-1.33
3	0.51	0.05	-0.34
5	0.62	0.26	-0.04
10	0.73	0.48	0.26

- After one year, only a record below -0.64 sends you away at 95%
- Time is the only lever that brings the line up to the goal

A manager is a strategy you cannot backtest

- Hiring is going in. Firing is walking away.
- The same three ingredients apply: a goal, a confidence, and time
- What is different: before you hire, you often have no returns at all, only what your selection process tells you

With a track record, the strategy logic applies

$$t \;=\; \widehat{AR}_T \times \sqrt{T}$$

- Three years at an appraisal ratio of 1: $t = 1.7$
- Ten years at 0.5: $t = 1.6$
- The record moves your belief before you have a single day of your own evidence

But the record you are shown was selected

- Of 100 managers with no skill, 5 will have a record that clears the 95% bar
- Those are the ones who come to see you

Their record is the **estimate** sample. The **test** starts the day you hire.

Now: the notebook

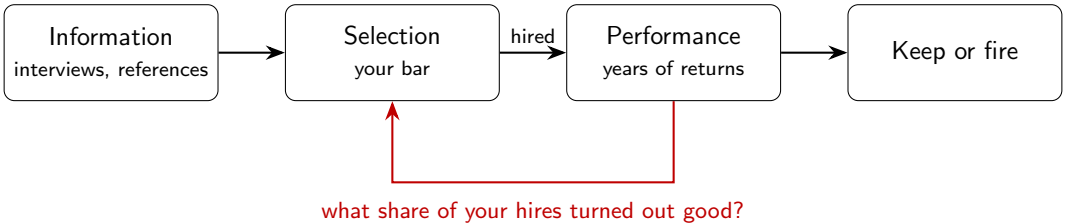
Notebook: Hiring and Firing — would you have hired Warren Buffett? And fired him?

1. **Would you hire him?** Slide through his record, 1989–1998, against the hire line.
2. **The test starts the day you hire.** You hire him in January 1999. Slide his record since then against the walk-away line.
3. **What happened next.** What firing him cost you.

Bring back: the rule only saw his returns. What did it not know?

Stop at **Back to the slides**; the challenge after it is homework.

The hiring process



- The walk-away rule sees only the third box. It does not know how the manager got there.
- Performance grades the manager, and it grades your process

No track record: you are betting on your process

- With no returns to look at, your belief that a hire is good is your hit rate
- The share of the people your process lets in who turn out good
- Returns then move that belief, one year at a time

The same bad years, two readings

- **Strong process**: most of your hires are good, so a bad stretch is probably bad luck
- **Weak process**: few of your hires are good, so a bad stretch is probably the truth
- The returns are the same. What differs is what you believed before you saw them.
- Melvin, January 2021: Citadel and Point72 read it the first way

How much two flat years should move you

Two kinds of manager: good (appraisal ratio a) or unskilled (zero). A T-year record is the truth plus noise, standard error $1/\sqrt{T}$.

$$\underbrace{\frac{\pi_T}{1-\pi_T}}_{\text{odds after the record}} = \underbrace{\frac{\pi}{1-\pi}}_{\text{odds at hire}} \times \underbrace{\exp\big(T\,a\,(\widehat{AR}_T - a/2)\big)}_{\text{how much likelier the record is if good}}$$

Hired to deliver $a = 1$, two years at zero: $\exp\big(2 \times 1 \times (0 - 0.5)\big) = 0.37$

share of your hires who are good	odds at hire	odds after ($\times 0.37$)	chance still good
80%	4.00	1.47	60%
50%	1.00	0.37	27%
20%	0.25	0.09	8%

How sure you must be to fire depends on your process

- Trust your process: demand more confidence before you fire
- Doubt your process: fire on less
- Two years in, hired to deliver 1: a strong process (80% good) fires only when 95% sure, a coin flip at 76%, a weak one (20%) at 39%

Firing means the returns have overturned what your process told you

The fire rule

What a hire costs, and what it pays

Your book has a Sharpe ratio of 0.5. You give the manager the risk their promise deserves.

hired to deliver	your Sharpe ratio if they are good	if they have no skill
1.0	1.12 (+0.62)	0.22 (−0.28)
0.5	0.71 (+0.21)	0.35 (−0.15)

- A manager with no skill adds risk and nothing else

Details

How long each one lasts

$$\text{hire if} \quad \underbrace{\pi \times \text{gain} \times \text{years the edge lasts}}_{\text{a good one}} > \underbrace{(1-\pi) \times \text{loss} \times \text{years until you fire}}_{\text{a bad one}}$$

hired to deliver	gain, lasting	loss, lasting	break-even share good
1.0	$+0.62$, 5 years	-0.28, 2.7 years	19%
0.5	$+0.21$, 5 years	-0.15, 10.8 years	60%

- Skill you can detect fast: one good hire in five is enough. Slow to detect: you need three in five.

Calibrating the bar

- **Nearly all your hires are good?** Your bar is too high. You are leaving talent out, and you can always fire the bad ones.
- **Fewer good than the break-even share?** Tighten the bar
- Your hit rate is an estimate too. Ten hires from a process that is right half the time can easily show 3 good ones, or 7

How do you know your hit rate?

You never see who is good, only records. Averaged over your past hires, the noise washes out:

$$\hat{\pi} \;=\; \frac{\text{average record of your past hires}}{a} \qquad\qquad SE(\hat{\pi}) \;=\; \frac{1}{\sqrt{N}}\sqrt{\pi(1-\pi)+\frac{1}{a^2\,T}}$$

	years of record per hire		
standard error of $\hat{\pi}$ (when $\hat{\pi}=0.6$, $a=1$)	1	2	5
10 hires	0.35	0.27	0.21
20 hires	0.25	0.19	0.15
50 hires	0.16	0.12	0.09

- Longer records shrink only the $1/(a^2T)$ term; more hires shrink both. Grade every hire over the same window, the fired ones included.

One decision problem

	a strategy	a manager
going in	record above $a^* + z_p/\sqrt{T}$	your process, plus any record
walking away	record below $a^* - z_p/\sqrt{T}$	the same test
your patience	confidence in the idea	confidence in your process

- Every exit is also evidence about how you entered

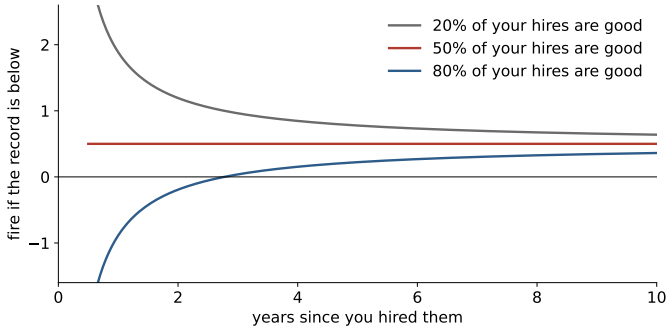
Homework: the challenge at the end of the notebook. The Buffett call again, this time with a prior.

Appendix

A1: When to fire

Fire once the manager is more likely unskilled than good: the odds after the record fall below 1.

$$\widehat{AR}_T < \frac{a}{2} - \frac{1}{a\,T} \ln \frac{\pi}{1-\pi}$$



Back

A2: The same rule, as a confidence level

Hired to deliver 1, two years in. When the record hits the fire line, how sure are you that the manager is below the goal?

share of your hires who are good	fire if the record is below	how sure you are by then
80%	-0.19	95%
50%	0.50	76%
20%	1.19	39%

- How sure: $\Phi\big((a - \text{line}) \times \sqrt{T}\big)$, the walk-away calculation run in reverse
- A stronger process sets a lower line, so you fire only at higher confidence

Back

A3: The Sharpe ratio arithmetic

- Your book alone has Sharpe ratio S. The manager is hired to deliver appraisal ratio a, uncorrelated with your book.
- The best mix gives the manager risk in proportion a/S to your book's
- **If they are good**: $\sqrt{S^2 + a^2}$
- **If they have no skill**: same risk, no extra return: $S^2/\sqrt{S^2 + a^2}$
- **Years until you fire**: $(1.64/a)^2$, to be 95% sure they are below the goal

Back

A4: Sources for the Melvin example

- **Annual and 2021 returns**, as reported in the press and compiled in Wikipedia, "Melvin Capital": 2015 +47%, 2017 +41%, 2019 +44%, 2020 +52% (2016 and 2018 not reported; about 30% a year 2014–2020). January 2021 −53%, February +22%, year to date −49% (March), −46% (June), −42% (November), −39% for 2021.
- **The injection.** WSJ, 24–25 Jan 2021: down about 30% so far in 2021; $2.75bn from Citadel and Point72 for money in the fund and non-controlling revenue shares. Benzinga, 25 Jan 2021: $2bn from Citadel and partners, $750m from Point72.
- **End of January.** Reuters, 1 Feb 2021: −53% in January; about $12.5bn at the start of the year, over $8bn at the end of January with the new money.
- **2022.** Reuters: −17% in the first three weeks of January 2022, −23% through April. CNBC: −21% in the first quarter.
- **Exits.** Bloomberg, 24 Feb 2022: Citadel pulling back most of its $2bn. Bloomberg, 12 Mar 2022: Point72 to redeem its $750m (a separate Point72 stake stayed). Melvin's letter, 18 May 2022: the funds will close.
- **GameStop prices**: Yahoo Finance, multiplied by 4 to undo the July 2022 split, so they match the prices traded in 2021. **S&P 500**: total return index (`^SP500TR`), Yahoo Finance, year end to year end.
- Between reported dates the fund's path is not known; the chart joins the reported points with dashed lines.

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